



Dynamic Voltage EV System (DVEVS): Achieving Enhanced Electric Vehicle Efficiency and Performance through a Switchable Series-Parallel Battery Configuration

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Abstract

The electric vehicle (EV) industry is relentlessly pursuing advancements to enhance efficiency, extend driving range and improve overall performance. A notable limitation of conventional EVs is their fixed-voltage battery architecture, which necessitates compromises in powertrain design and can result in suboptimal efficiency under varying driving conditions. The Dynamic Voltage EV System (DVEVS) presents a transformative solution, allowing the battery pack to dynamically reconfigure its cells between series and parallel connections. This review explores the core principles of DVEVS, including battery topology,

power-electronics-based switching, and the integration of hybrid energy storage solutions such as electric double-layer capacitors (EDLCs).

We explore the foundational concepts of battery reconfiguration, delve into specific implementation strategies such as power-electronics-based switching and hybrid energy storage systems and address the critical need for adaptive thermal management and advanced charging infrastructure. This review synthesizes a holistic understanding of the DVEVS concept as a transformative approach to achieving reliability, adoptability along with greater efficiency and promising future research in next generations of electric mobility

1. Introduction

The efficiency in electric vehicle is a key factor; thus automobile industry is striving towards advancing in it such as extended driving range and improvement in performance. The study here investigates on battery electric vehicle which is always segmented by range or performance. Powertrain design also focuses on this aspect by choosing efficiency or performance. Themselves as two different entities can be switched in single vehicle which is defined as one entity.

As battery technology evolving, competition in automotive world is in peak. Demands on key factors like charging time in EV is a significant value. This allows us exploration on potential architectural changes in battery modules. Conventional EV battery packs are designed with fixed number of cells in series or parallel resulting to static nominal voltage. High voltage is advantageous for high-speed cruising in turn Improving efficiency by reducing current and associated resistivity losses. On other side lower voltage is beneficial for rapid acceleration and delivering the high torque as needed.

Dynamic voltage electric vehicle system now onwards DVEVS; offers complete shift to the conventional approach of static nominal voltage incorporating both series and parallel connection making switchable battery configuration.

The terminal voltage can be altered in a real-time allowing adaptability in voltage allow powertrain to operate in peak efficiency with wider range of speed and torque. Furthermore, results to energy economy and peak vehicle performance. The whole system also has profound impact on thermal management of battery pack not just limiting to electrical efficiency. Reconfigurable voltage system can enhance reliability along, intelligently managing individual cell modules for both electrical and Thermal aspects. The study also includes and explore the foundational principles. Implementation strategies and findings in the literature [1, 2, 3, 4, 5] and furthermore lights on the charging infrastructure requirements of DVEVS.

2. Dynamic Voltage Electric Vehicle System (DVEVS)

DVEVS lies in the ability to manipulate the battery configurations pertaining on the voltage between the constituent battery modules, considering ideal cell to cell variations in capacity and resistance. The whole system incorporates switching mechanisms which includes Contactors, MOSFETs, SiC switches and power electronics between

the cell modules. This becomes a new complex technology with various advantages and disadvantages. This system constitutes of:

1. Cell topology / Cell Connection
2. Electromechanical switching Systems
3. Battery Thermal Management System (BTMS)
4. Charging infrastructure

These characteristics are explored in detail, this also concluded by dissecting each in discussions.

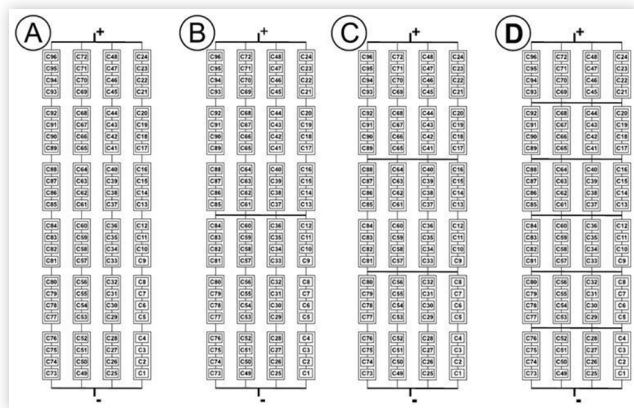
Battery architecture is often classified as cell-to-module, cell-to-pack, or cell-to-chassis. The mechanical integrity of the battery structure varies depending on the chosen architecture.

2.1. Series Connections

The connecting cell modules in series, where positive end of cell terminal connected to negative end and connecting it further to their neighboring cells. This will increase total voltage in the battery cells, but Current rating of battery remains constant. Refer A in Figure 1

- Helps in attaining Higher voltage and lower current for same power output
- This is very beneficial in achieving higher motor efficiency at high speeds

FIGURE 1 Studied topologies.[2]



- In series string the performance of entire configuration is detected by weakest module resulting under performance.
- An underperformance or faulty cell or cell module in the long run can limit the usable capacity of entire series chain or string.

2.2. Parallel Connections

The Connecting cell modules in parallel where negative end and positive end respectively connect to other cell modules. This amplification with cells will increase the

total current rating of the battery pack but the voltage of battery remains same as that of single module. Refer D Figure 1

- This is very beneficial in achieving the acceleration providing high current.
- In parallel connection the system offers redundancy which in turn continues to provide power supply even in case of failure in cell module.

2.3. Cell Topology

here are different topologies to achieve in robustness to variation occurring between the cells often refer to cell mismatch one of the studies [2] shows below character, Figure 1

- Parallel cell module [PCM] configuration. Here the cells are connected in parallel form, modules are later connected in series.
- Series cell module [SCM] configuration, here the cells are connected in series form, modules are later connected in parallel form.

PCM Offers more resilient to capacity variations. SCM offers more self-balancing within cell modules These battery configurations and topology clearly exhibit the advantages and further motivate them to develop the DVEVS. The system itself harnesses the benefits of both configurations and topology while a sophisticated and intelligent battery management system can mitigate the drawbacks associated with cell topology.

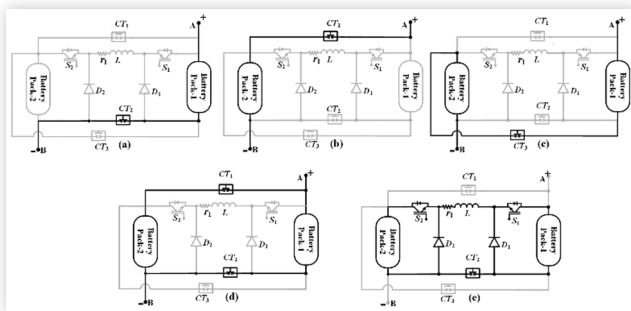
3. Power Electronics-Based Reconfiguration

To realise the DVEVS strategy, [3] showcased utilization of power electronics for configuring the cells in series and parallel. The research highlights circuit configurations which are based on contactor, which acts like switch and could be used to configure circuit dynamically for achieving different purpose(s). There are five different operating modes proposed, which utilizes two battery pack as power source and are highlighted as follows:

- **Both Battery Packs in series:** This mode connects both battery in series increase in the voltage.
- **Both Battery Packs connected in parallel:** This mode connects pack in parallel which increases current rating.
- **Battery Pack-1/Battery pack-2:** This Connects either of the battery pack directly to load.
- **Voltage balancing mode:** This is very similar to parallel mode but rather than connecting battery to load, it disconnects the load from circuit.

When battery is connected in series, the total voltage increases. This configuring helps in reducing I^2R losses, which makes an ideal scenario for charging. When the same is connected in parallel the current rating increases. This condition is useful when small burst of current is required such as acceleration in vehicle. On contrary, as voltage reduces the current flow increases making current carrying components heavier and bulkier. The third configuration is advantageous in a peculiar scenario, when either one of the battery is fails to work, other battery acts as a redundancy. This configuration can be used when demand of the load is less. Each battery is built differently SOC can be balanced using the voltage balancing mode. This helps mitigation of circulating currents.

FIGURE 2 Different operating modes. [3]

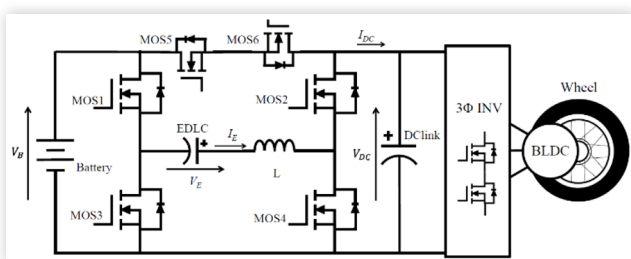


4. Hybrid Energy Changeover Systems

This novel approach mentions the literature by [4] integrates two distinct system, battery and electric double layer capacitor (EDLC). The combined duo system works together, combines advantages of battery which is main source of energy and high in energy density. Along with EDLC which is high power density source having an edge in events such as Acceleration and deceleration.

Additionally, the literature also showcases a system which can connect these two either in series or in parallel. When the system is in parallel mode, if EDLC has sufficient charge, along with battery this configuration works together to power motor. When EDLC voltage reduces below battery voltage system changes to series mode extracting all power available in EDLC.

FIGURE 3 series or parallel changeover system [4]

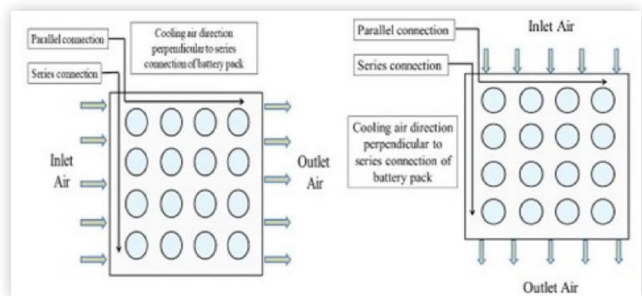


A Combined symphony of two system. increases performance and extend battery life by unloading peak power requirements to EDLC. One of the downside of such configurations is managing two different storage technologies making it complex and expensive.

5. Thermal Management

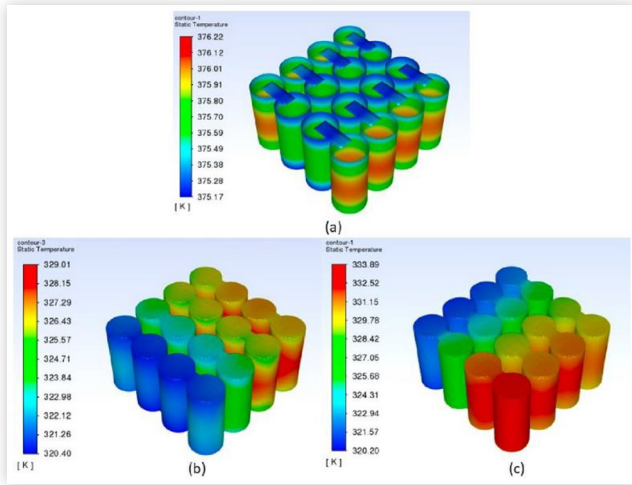
DVEVS system will have a large effect on thermal profiles due to its dynamic nature of switching cell modules. This itself is a challenge and also raises the opportunity and need for advanced thermal management. The critical insights on the thermal behavior of different topologies and its thermal behavior can be largely seen by research reference [1] The fundamental relation ($P = VI$) relates the thermal behavior of the different cell topologies along with impact from busbar connection [1] The heat generation is directly proportional to I^2R where Heat is $H = I^2Rt$. This clearly shows us parallel connection will generate significantly more heat, since delivering the same power requires a much higher current. Where in series connection voltage will be increased making less resistance towards delivering current in turn heat generation is very little

FIGURE 4 Airflow strategies [1]



Busbar is very crucial to analyze in DVEVS system. The requirement of transferring higher power by busbar moreover the increase in contact resistance eventually may turn into thermal runaway and short circuit. The testing result shows the actual current which busbars carries will affect the battery pack [1] the work here can give insights on the material thickness and length of busbar which effects the cell topology scenario.

FIGURE 5 Comparison of cooling efficiency for different airflow strategies [1]

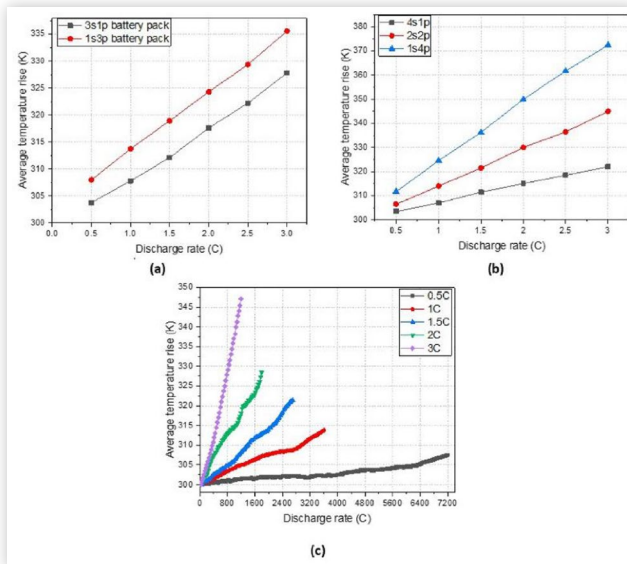


Cells in parallel connection share total current which also led to change in ohmic resistance and unequal current distribution. Cells with lower resistance can carry more current, causing localized warming zones. This can cause thermal gradients in certain locations and further poor heat dissipation can cause temperature increment. A cell must operate at temperatures ranging from 20 °C to 45 °C to have a higher life cycle and lower degradation of electrolyte.

5.1. Targeted Thermal Management

This behavior from cells on DVEVS leads to the need for dynamic thermal management. The system should be intelligent enough to direct the cooling liquid as cell temperature is out of specification. Especially parallel connection requires more effective cooling as parallel connections are more prone to hotspot. (BMS) battery management system and Battery Thermal management system (BTMS) should go hand in hand without delay as DVEVS system switches as in when required.

FIGURE 6 Impact of battery connection [1]



Cooling requirements are much reduced in high-speed cruising mode as series configuration operates with less resistance. Thus, saving total energy.

6. Charging Infrastructure

Electric Vehicle Supply Equipment (EVSE) is the most critical feature of any Electric Vehicle (EV) system. This majorly requires faster energy filling from grid to vehicle battery. DVEVS extends to charging process optimization rather than only concentrating on driving performance.

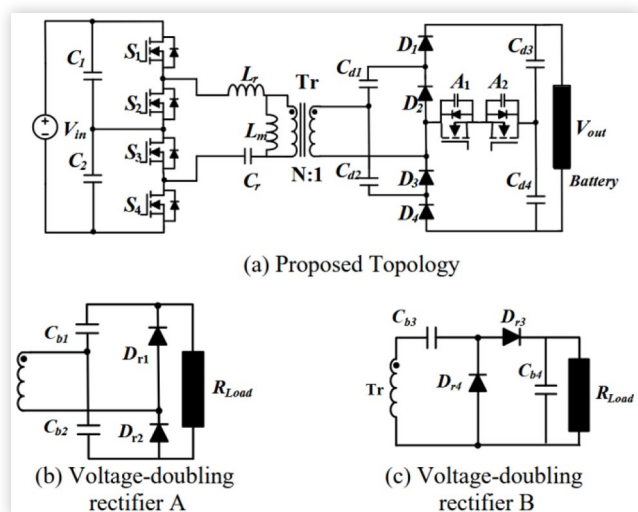
Battery pack reconfigured into a high voltage series connection during charging inducing much faster charging performance with lower current. This also makes it much more efficient in thermal losses. Apart from that it also accommodates a wider range of voltage for charging making wide acceptance of charging systems in EV. The research from [5] shows a new methodology called reconfigurable LLC resonant converter.

Due to DVEVS system the voltage range of charger is efficient and operate with broader output from 200V to 800V. A rectifier is employed, can function as double voltage or quadruple voltage rectifier depending on EV battery voltage requirement. This allows the charger to maintain high efficiency, especially not limiting to higher voltage or low voltage.

Voltage stress is another factor to be broadly investigated for electric components basically enhancing reliability of the charging station [5]. DVEVS can be utilized for full potential of charging without loss in current as dynamic charging facilities are available in market. Current study by [5] demonstrates a rate of operating with 400 V input and output range of 200-800V at rated power of 500W with peak.

efficiency up to 93.9%. This soft switching can be dynamically changed across the range in DC charging cycle.

FIGURE 7 Reconfigurable LLC Resonant converter [5]



7. Discussion

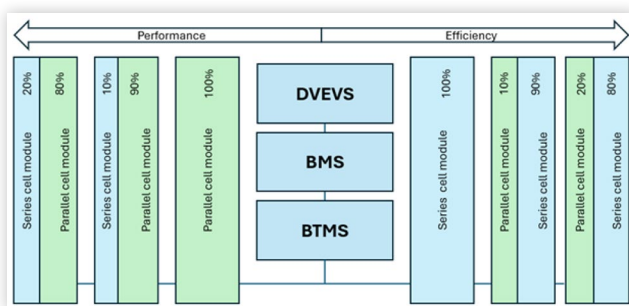
The switching in battery offers advantage either resilience in capacity or redundancy using self-balancing. This allows parallel circuit to impact on efficiency and performance where series leads to more efficient and power usage from battery. Further steps clearly show us need for Configurations in automotive industry. The design A, SCM and D, PCM are at extremities. B and C with hybridization can make better choice in negotiating advantages either on SCM or PCM and switch between hybridization's mode. [2]

It is very important to choose between the topologies and its requirements considering array of series and parallel connection with different nominal voltage and capacity. The choice here is made for requirement of flexibility but with constant total power output.

- An ideal topology for motor efficiency can be hybrid circuit design, considering SCM as base with hybrid parallel connection on top (Series connection as upper hand).
- But for performance and initial acceleration the hybrid circuit design, considering PCM as base with hybrid series connection (parallel connection as upper hand).

DVEVS can allow alternative path in the case of cell failure. Making vehicle run even during this scenario. where the failure in one cell can lead to have other cell still supply power. DVEVS gives a major opportunity to provide Parallel architecture and topology to run rather than complete breakdown in fixed topology condition.

FIGURE 8 Hybrid Series-Parallel System



7.1. Thermal Management

BTMS and BMS are major part in DVEVS system. The system becomes complex, when regulating temperature, battery health and rapid adjusting to environmental condition. BTMS must Analyse the temperature rise for battery packs especially, in parallel connection, which is crucial as contact resistance is at peak. Thus, affecting the overall cell performance and causing battery degradation. The System should take this scenario and bring topological order from PCM to SCM though coordination from BMS and BTMS.

7.2. Integrating Diverse Technologies

Considering outlook, the energy density of automobile battery is projected to increase. The increase in the capacity demands usage of multiple cells/modules/battery for different automobile electric requirement. The electric requirement being different for each application (such as, driving, Low Voltage Electronics, HVAC, etc), the need for different voltage will arise. But on the contrary, Charging the battery will always require high voltage to reduce ohmic losses. Hence need for DVEVS is and corresponding convergence of novel switching technology is inevitable. Power electronics plays important role in making such system reality, it can be readily concluded that, just considering one such technology [4] will not suffice the requirement. The power electronics has many advantages as evident in previous discussion also capable of achieving wide range of Voltage to cater the applications. But they also have fair share of limitation one of being ability to carry switching for high and very high current rating. This is due the fact that power electronics suffers with high ohmic losses which increases with a function of square with every increase in the current.

It is need of hour to maximize the utilization of energy and deliver to required application, EDLC is being used that not only supports the battery power application but also go hand in hand for better user experience. This technology can be demonstrated using 48V battery which is extensively used in hybrid powertrain. Another diversification can be demonstrated along with Hydrogen cell. Hydrogen cell being far superior technology in terms of energy density along with faster refuelling capability can prove beneficial in heavy automobiles. Considering vehicles combining technologies like batteries, fuel cell or high-density capacitors bank which works in tandem taking the strength of each other is not farfetched.

7.3. Limitation in Switching Mechanisms

The ability to reconfigure battery modules in real-time between series and parallel arrangements delivers notable performance benefits in dynamic voltage electric vehicles system (DVEVS). However, this dynamic switching introduces a potential delay like the hesitation experiences when shifting gears in a manual transmission vehicle. Just as a poorly timed gear shift can cause a temporary loss of power or responsiveness, a delay in switching battery configurations can result in torque lag or reduced acceleration.

This delay primarily affected by the type of switching technology used. For instance, Solid-state relays (SSRs), utilizing components like MOSFETs, IGBTs, or Silicon Carbide (SiC) transistors, offer microsecond-level switching and are well suited for underload transitions, due to their current operation demands sophisticated thermal management.

In contrast, electromechanical contactors are robust and capable of handling high voltage and currents but suffer from slower switching time(10-50ms) and mechanical wear, making it more suitable for transitions during idle or low-load.

Hybrid Switching, combining solid state relays (SSRs) and contactors, strikes a balance by offering fast response times along with robust current-handling capabilities. However, this configuration introduces additional complexity in control algorithms.

The most advanced approach lies in integrated power modules that consolidate sensors, microcontrollers and switching elements into compact units. Capable of real-time diagnostics and adoptive switching based on state-of-charge (SOC), temperature and real time driving demands.

On the charging side, DVEVS architecture utilizes reconfigurable LLC resonant converters and multi-mode rectifiers to accommodate a wide voltage range(200V-800V), enabling efficient high-voltage charging. Despite their benefits, scaling such systems presents challenges in cost, voltage stress management and standardization. Addressing this issue requires holistic integration of power electronics, thermal management and intelligent control systems to ensure seamless operation without compromising vehicle performance.

7.4. Charging Infrastructure for DVEVS

The DVEVS concept offers groundbreaking advancement in electric vehicle battery architecture by enabling real time reconfiguration between series and parallel connection. This flexibility allows for optimized performance while driving and efficient charging. However, the full benefits of DVEVS can only be realized if charging infrastructure evolves in parallel to support its wide and variable voltage range(200V-800V). This creates a classic chicken and egg problem

- Without widespread infrastructure (example: chargers with reconfigurable LLC resonant converters and adaptive rectifiers) [5], manufacturers may hesitate to adopt DVEVS at scale.
- Without mass adoption of DVEVS equipped vehicles, infrastructure providers may hesitate to invest in upgrading charging stations until DVEVS become more common on the road.

This mutual dependency slows the pace of innovation and market development. For example, DVEVS system can switch to high-voltage series mode during charging to reduce the current and thermal losses, yet this capability is only effective if the charging station can handle variable voltage inputs and outputs something not yet standard in most public or residential charges. Similarly, the power electronics and switching system such as solid-state relays (SSRs) and hybrid switches needed for seamless driving performance are complex and costly,

and their benefits are maximized only when supported by compatible infrastructure.

Thus, the chicken and egg scenario in DVEVS reflects the need for synchronized development in both vehicle technology and charging infrastructure. Breaking the cycle may require standardization of voltage interfaces, government incentives or pilot programs that demonstrate DVEVS benefits in real world conditions.

7.5. Policies and Standardization

The Dynamic Voltage EV System (DVEVS) with switchable series-parallel battery configurations offers a transformative approach to enhancing electric vehicle efficiency and performance.

Successful implementation requires comprehensive policies and standardization efforts. This outlines key areas for consideration to guide industry stakeholders, policymakers and researchers.

- Establish rigorous safety protocols for battery management and voltage regulation to ensure reliable and safe operation.
- Create standards for interoperability between different manufacturers, DVEVS components. Ensuring compatibility between ECUs, charging systems will promote widespread adoption and ease integration.
- Standardize communication protocols for real-time data exchange between DVEVS components to enable seamless operation and diagnostics, enhancing system efficiency and troubleshooting.
- Foster collaboration between automakers, technology providers and regulatory bodies to align goals and streamline DVEVS implementation. Joint efforts will facilitate knowledge sharing and accelerate progress.

7.6. Charging Communication and Protocols for EVSEs

Different technologies being fore front of the EVs, to utilize each to fullest, technologies need to have good communication. The communications flexibility between EVSEs in catering different EV platform requirements plays important role in acceptability of EVs. This also opens possibility of having a standardization of protocol for DVEVS scenario. Thus, future amendments and extension will be required for communication laid by ISO 15118. For instance, EVSE will start charging with high voltage series connection initially. If there BTMS detects higher temperature of battery, it could request EVSE to reduce the voltage which in turn reconfigures the battery thus, charging battery by any eliminating losses arising from any DC-DC converters, making the charging dynamic utilizing the system to maximum.

7.7. Decentralized Electronic Control Units (ECUs)

Decentralized Electronic Control Units (ECUs) in electric vehicles (EVs) are future technology where automobile industry are looking forward, this offers significant advantages for dynamic voltage systems. This approach leverages advanced chip technologies to enhance performance, reduce costs and simplify system architecture.

Decentralized ECUs enable precise energy management by dynamically adjusting voltage and optimizing charging cycles in real-time. This leads to more efficient charging and extended battery life.

Localized control allows each ECU to act as a mini-brain and manage specific components like battery, motor, auxiliary systems etc. This ensures every part of the EV operates at its best, providing a smoother and more responsive driving experience.

Advanced chip technologies offer the processing power to handle complex tasks and integrate multiple functions. This improves overall performance and streamlines the system, making it more efficient and less prone to errors.

Decentralized ECUs also simplify fault isolation, making it easier to identify and fix issues, thereby enhancing reliability. They offer greater flexibility to adapt to future technological changes and upgrades, ensuring the vehicle remains at the cutting edge of innovation. In practice this could mean dedicated ECUs for battery management, motor control and auxiliary systems work in synchronization to optimize energy use and enhance vehicle performance.

By adopting decentralized ECUs and advanced chip technologies, electric vehicles are not just receiving an upgrade, but they are being propelled into a new era of efficiency, affordability and adaptability.

7.8. Cell Level Reconfiguration

The success of on-board Dynamic Voltage EV Systems (DVEVS) relies on developing efficient, compact and cost-effective switching components and power converters. Research into wide-bandgap semiconductors like silicon carbide (SiC) and gallium nitride (GaN) is crucial to minimizing size and losses.

To ensure seamless and safe charging across different manufacturers, standardizing communication protocols between the vehicle's Battery Management System (BMS) and charging stations is essential.

Long-term studies are necessary to validate the reliability of switching hardware and quantify real-world gains in battery lifespan, considering both electrical and thermal cycling stresses.

Future systems could extend reconfiguration to the individual cell level, offering ultimate flexibility in battery pack management, isolating faulty cells and maximizing usable capacity and lifespan.

8. Conclusion

The Dynamic Voltage EV System (DVEVS) represents a significant leap forward from the conventional fixed-voltage architecture of electric vehicles. By enabling the battery pack to dynamically reconfigure between series and parallel connections, DVEVS enhances powertrain efficiency, improves performance and effectively manages thermal loads to extend battery life. The strategies reviewed in this paper from on-board power electronics and hybrid storage systems to adaptive thermal management and flexible charging infrastructure highlight the multifaceted nature of this technology.

Switch Type	Technology	Advantages	Challenges	Use Case in DVEVS
Solid-State Relays (SSRs)	MOSFETs, IGBTs, or SiC transistors	Fast switching, no mechanical wear, integrated control logic	Heat dissipation, cost at high current levels	Ideal for frequent switching between series and parallel modes under load
Electromechanical Contactors	Mechanical switches rated for HV and high current	Proven reliability, high current handling	Slower switching, mechanical wear, arcing risk	Suitable for switching during idle or low-load conditions
Hybrid Switches	Combination of SSRs and contactors	Balances speed and robustness, reduces arcing and wear	Complex control, higher cost	Optimal for dynamic systems requiring both performance and safety
Integrated Power Modules	Custom modules with switches, sensors, microcontrollers	Compact, intelligent, realtime diagnostics	Design complexity requires robust firmware	Enables smart switching based on SOC, temperature, and driving conditions

While challenges related to complexity, cost, and control remain, the potential benefits are compelling. The future of EV technology will likely involve a holistic approach, where intelligent, reconfigurable vehicles with integrated electrical and thermal management systems communicate seamlessly with equally intelligent charging infrastructure. As research continues to advance the efficiency and reliability of the underlying components and control systems, DVEVS is poised to revolutionize the electric vehicle industry.

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Definitions, Acronyms, Abbreviations

BMS - Battery Management System

BTMS - Battery Thermal Management System

DVEVS - Dynamic Voltage EV System

EDLC - Electric Double Layer Capacitor

EV - Electric Vehicle

PCM - Parallel-Cell-Module

SCM - Series-Cell-Module

ECU - Electronic Control Unit

SSR - Solid-State Relays

SOC - State of Charge

EVSE - Electric Vehicle Supply Equipment

MOSFET - Metal-Oxide-Semiconductor Field-Effect Transistor